

4-assignment-esm-262

AUTHOR

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Building your first function

Defining contract

This function calculates Simpson's Index.

Input: Number of individuals per species and total individuals of all species

Output: Simpson's Index (D), value between 0 to 1

The value represents the probability that 2 individuals randomly selected from a sample will be the same species.

Parameters:

species, the number of individuals per species

total, the total of all individuals across species

Return:

index, the output of Simpson's

Function: The function will be the sum of each individual species total ("species") divided by the total of all species ("total") squared.

$$D = \sum((\text{species} / \text{total})^2)$$

```
#' @param species individuals per species  
#' @param total sum of all individuals across all species  
#' @return index (D)
```

Function Definition

```
# function definition  
index = function(species, total = sum(species)) {
```

```
# here's where I calculate Simpson's index
result = sum((species/total)^2)
return(result)
}
```

Testing function with example data

```
# generate test data frame
test_data_s1 <- data.frame(
  species_name = c("redleg","arroyo","yellowleg","catree","baja")
  individuals = sample(1:50, size = 12)
)

test_data_s2 <- data.frame(
  species_name = c("redleg","arroyo","yellowleg","catree","baja")
  individuals = sample(0:20, size = 12)
)

# run data in function

site1 <- index(species = test_data_s1$individuals)

site2 <- index(species = test_data_s2$individuals)

site1
```

```
[1] 0.1141242
```

```
site2
```

```
[1] 0.119049
```